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One input, one doubled result.

Five floating-point inputs stay in the same order.

> Explicit teaching thread mapping · no GPU timing claim

Five FP32 inputs → five doubled outputs

Array index	Input	Output
0	1.0	2.0
1	2.0	4.0
2	3.0	6.0
3	4.0	8.0
4	5.0	10.0

Same calculation · PyTorch chooses its implementation.

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A thread executes the GPU program.

Our mapping gives one input position to each thread.

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One input position per participating thread

Array index	Input	Output
0	1.0	2.0
1	2.0	4.0
2	3.0	6.0
3	4.0	8.0
4	5.0	10.0

Index 4 selects value 5.0 and writes 10.0.

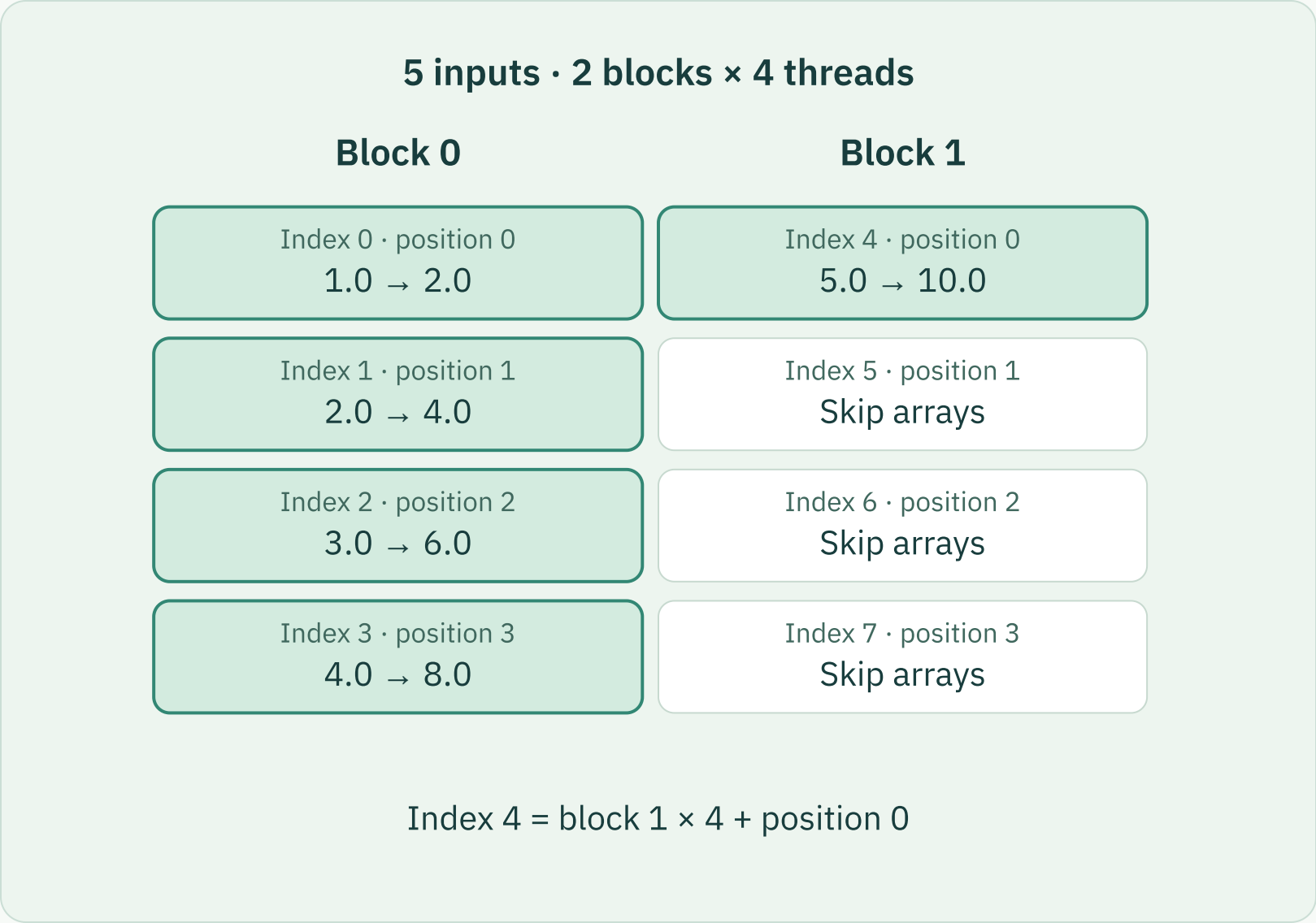
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Blocks group the threads.

The global index combines block and local position.

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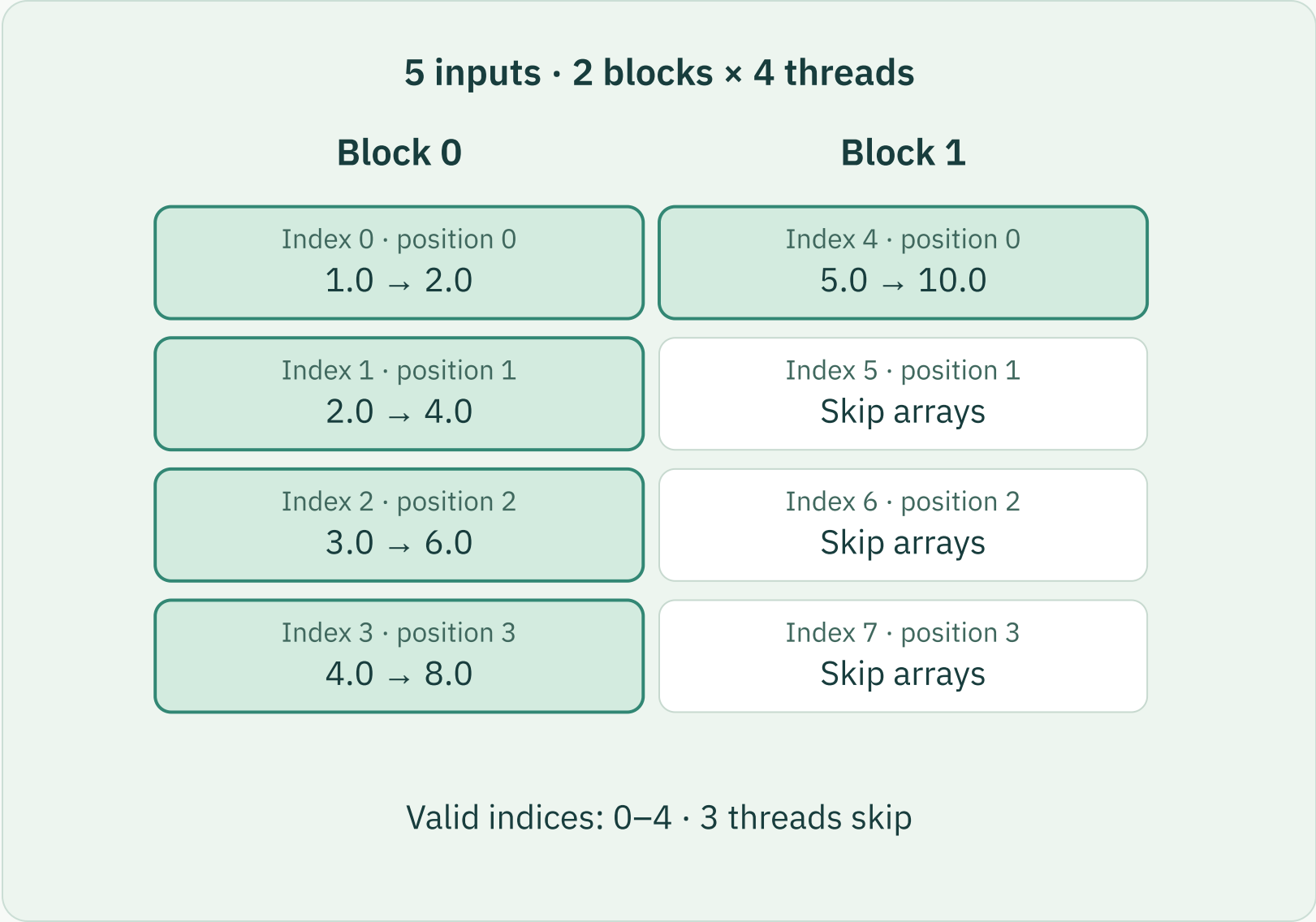
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The CPU supplies the launch and count.

Launch coverage and valid array positions are different.

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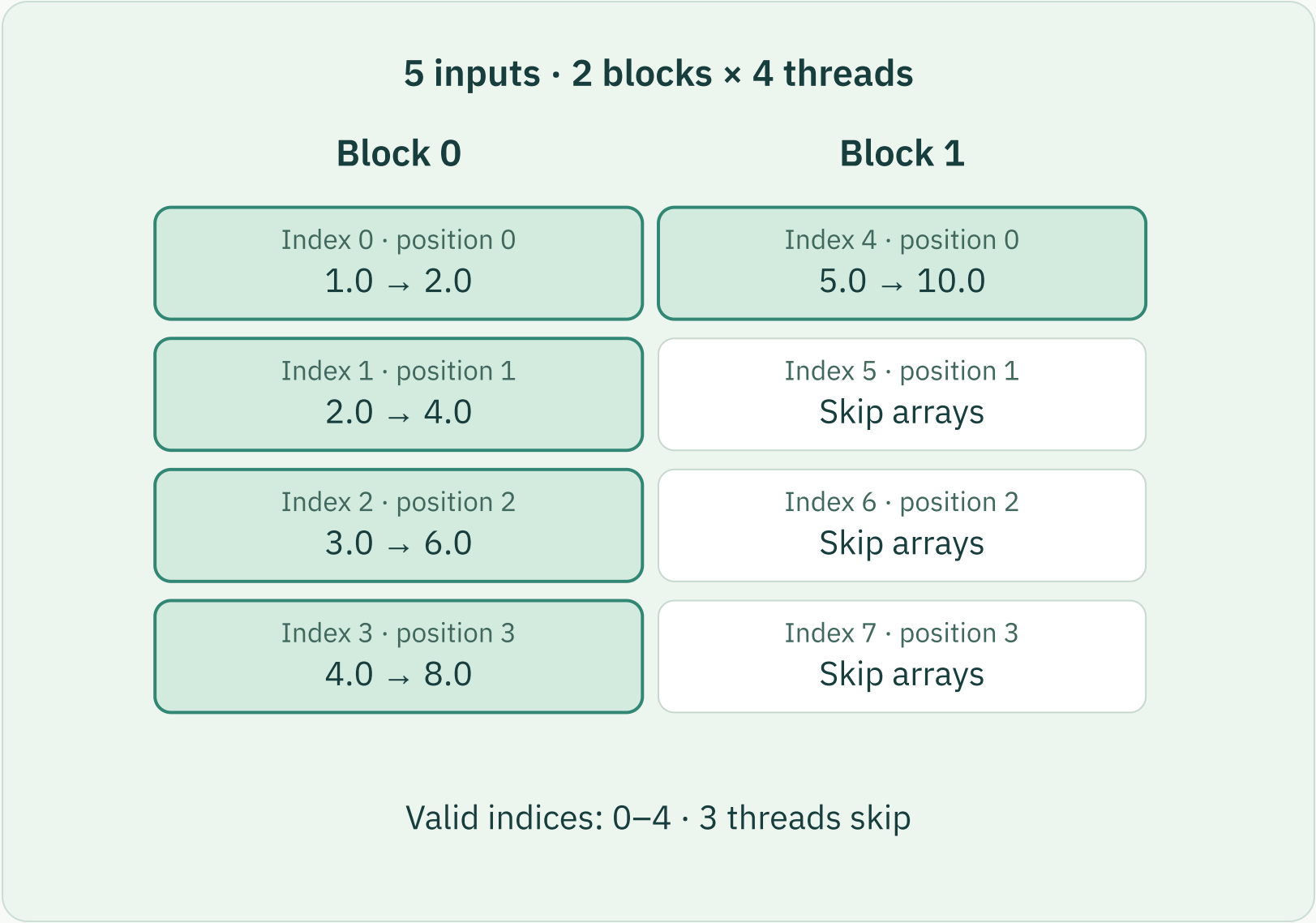
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A bounds check protects the arrays.

Only indices smaller than the supplied count may access them.

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Pause: cover nine inputs.

Keep four threads per block.
Choose the block count.

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Nine inputs · four threads per block

1.0	2.0	3.0	4.0
5.0	6.0	7.0	8.0
9.0			

Two blocks × four threads = eight threads

Does every input have a thread?

Smallest block count: ?

Extra indices that must skip: ?

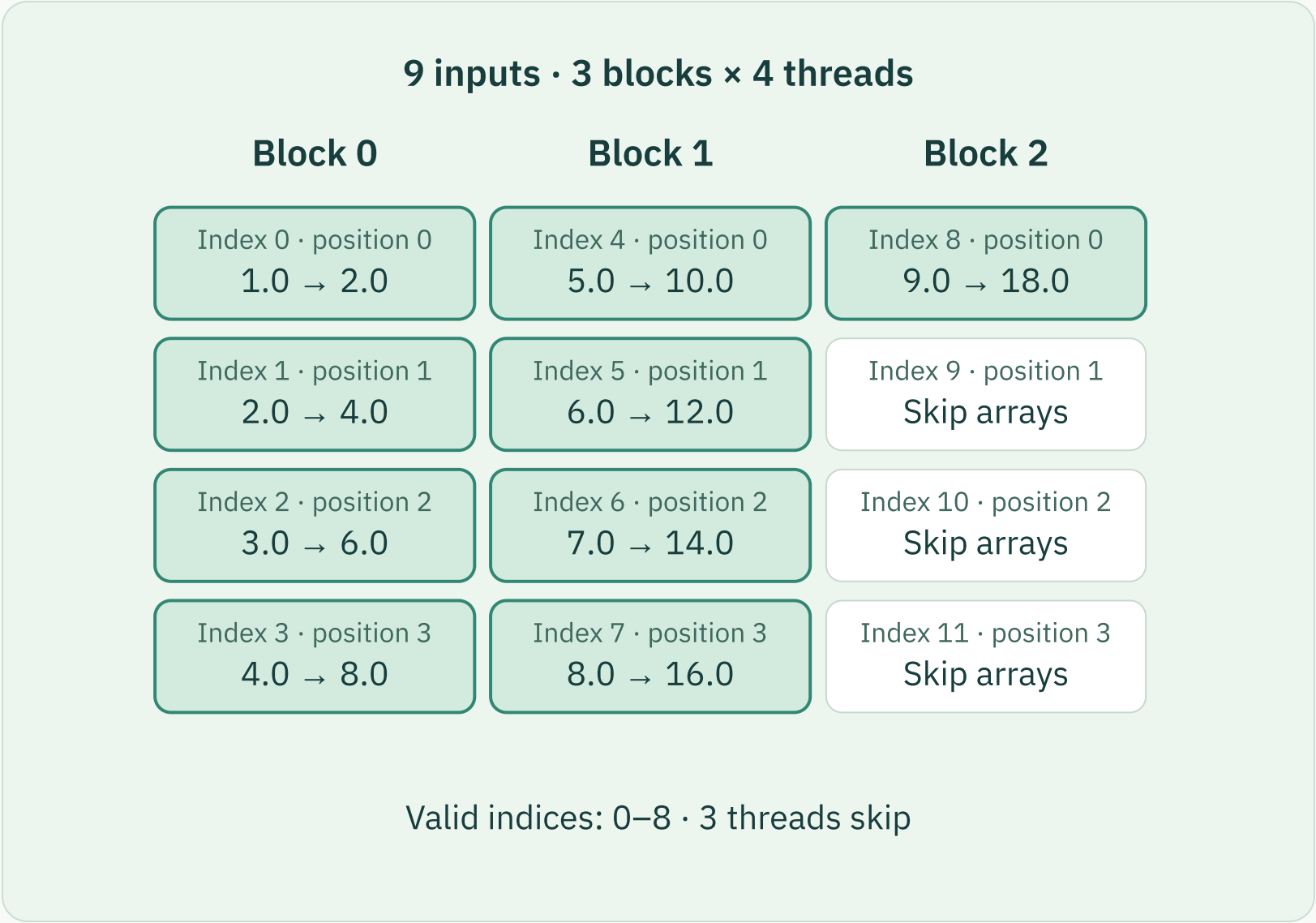
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Three blocks cover all nine inputs.

Twelve launched threads include three extra indices.

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Trace values, indices and coverage.

Return to the chapter and explain a different grouping.

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Explain three different quantities

Array index
Which input and output does this thread use?

Launch coverage
How many threads do the requested blocks create?

Bounds check
Which of those indices may access the arrays?

Changing the grouping does not change doubling.

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